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Smart Healthcare Robot

Bringing Care Closer to Patients in Oman

Final Year Project Report

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1
Introduction

This chapter establishes the project context, problem statement, motivation, objectives, methodology overview, and the project management plan that frames execution for the rest of the report.

1 Introduction

1.1 Context

Healthcare in Oman faces a simple but serious challenge: doctors cannot always be where patients needthem. This affects everyone - elderly people who cannot travel to hospitals, children in rural areaswho need specialist care, and patients in remote communities where the nearest doctor might be hoursaway. While Oman's healthcare system reaches 100% of urban families, 10% of rural families still lackadequate access to medical care.When doctors are unavailable due to emergencies, travel, or simply being in another location, patients wait. Sometimes they wait too long. This is particularly difficult for elderly patients who mayhave mobility issues, and for children who need comfort and reassurance during medical situations.2

1.2 Problem Statement

Oman's healthcare system strives to reach all citizens, with government local health services reaching 100% of urban families and 90% of rural families [1], yet geographic distances and resource constraints create barriers to timely and accessible care for the remaining underserved populations. Existing telepresence solutions often rely on manual control and lack the ability to engage patients in even basic diagnostic dialogue. This results in increased workloads for healthcare providers and a less personalized, less effective experience for patients, particularly those in rural areas or with mobility limitations.

1.3 Motivation

Every day in Oman, patients need medical attention that could be provided remotely if the right tools existed. Children in hospitals need comfort when parents cannot be present. Elderly patients need medication reminders and someone to check on their wellbeing. Rural communities need access to specialist consultations without long journeys.

The global demand for remote healthcare solutions is growing rapidly, with the telepresence robot market expected to reach significant growth. More importantly, Oman's Vision 2040 calls for healthcare innovation that serves all citizens equally, regardless of location [2]. Recent studies have shown that telemedicine adoption in Oman's primary healthcare settings demonstrates both promise and areas for improvement [3], while infrastructure requirements for effective telemedicine networks continue to be refined [4]. The aim of the project is to design and implement a smart telepresence robot with autonomous indoor navigation and on-device basic triage dialogue for ward corridors and bedside spaces in hospitals in Oman.

1.4 Objectives

Our objectives focus on solving real problems for real people:

- **To develop a robust indoor telepresence robot with autonomous navigation**
- **To engineer a safety-aware navigation system for crowded areas**
- **To design a clean, user-friendly system with an intuitive graphical user interface (GUI)**
- **Secure, real-time A/V consult suitable for bedside use.**
- **A pilotable demo scenario (nurse station → Bed X → short consult → return)**

1.5 Methodology

The project will begin with a user-centric research phase, including interviews with Omani doctors to define precise requirements, operational constraints, and key interaction preferences. Following this, we will evaluate and select optimal design alternatives for both the hardware platform and software architecture and proceed with an iterative prototyping and development cycle. This phased approach will begin with establishing reliable remote tele-operation, advance to implementing and refining the autonomous navigation system, and culminate in integrating the on-device dialogue capabilities. Finally, the system will undergo real-world validation in simulated and actual settings to measure performance metrics such as navigation reliability, quality of service for remote consultations, and overall experience for both patients and clinicians.

1.6 Project Management Plan

1.7 Timeline and Work Breakdown Structure (WBS)

The project is planned over a 12-month period. The major tasks, durations, dependencies, and deliverables are outlined in the Work Breakdown Structure presented in Table 1.

Table 1: Work Breakdown Structure (WBS)

ID	Task Description	Duration (weeks)	Deliverables/Checkpoints	Assigned Member
1.0	**Phase 1: Research & Setup**			
1.1	Literature Review & Requirements Finalization	4	Requirements Document	-
1.2	Hardware Procurement & Assembly	2	Assembled Robot Platform	Almoulla
1.3	Development Environment Setup	2	ROS & Cloud System Configured	Abdulmunim
2.0	**Phase 2: Core Development**			
2.1	Basic Navigation in Simulation (navigation)	2	Successful Robot Motion in Gazebo/Sim	
2.2	User Interface (UI) Initial Design	1	UI Mockups & Android Project Setup	Abdulmunim
3.0	**Phase 3: Integration & Features**			
3.1	Initial Hardware/Software Integration	1	Basic Tele-operation on Hardware	Almoulla
3.2	Navigation Refinement (Hardware Test)	2	Autonomous Navigation on Robot	Almoulla
3.3	LLM Integration (Core Dialogue)	2	Basic Virtual Nurse Dialogue Integrated	Almoulla
3.4	UI Development (Core Features)	2	Functional Clinician/Patient GUIs	Abdulmunim
4.0	**Phase 4: Testing & Refinement**			
4.1	Integrated System Testing (Lab)	1	Initial Bug Report	-

Work Breakdown Structure (WBS) (Continued)

ID	Task Description	Duration (Weeks)	Deliverables/Checkpoints	Assigned Member
4.2	LLM Dialogue Refinement	1	Improved Dialogue Responses	-
4.3	UI/UX Testing and Improvement	1	Enhanced User Interface	-
4.4	Navigation Stress Testing	1	Performance in Varied Conditions	-
5.0	**Phase 5: Finalization & Reporting**			
5.1	Final Prototype Assembly	1	Fully Assembled and Functional Prototype	-
5.2	Comprehensive Evaluation	1	Final Evaluation Report, User Feedback	-

2

Literature Review

This chapter surveys consumer, industrial, and healthcare robotics; documents benefits and challenges of AMRs in hospitals; analyzes core technologies (navigation, RF localization, SLAM, LLMs); and motivates the platform choices and project needs.

2 Literature Review

2.1 Motivation and State of the Field

The integration of robotics and artificial intelligence into healthcare is advancing rapidly, driven by challenges such as aging populations, rising healthcare costs, and medical staff shortages. Researchers have developed various robotic solutions, including telepresence robots for remote consultations, autonomous robots for logistics, and socially assistive robots (SARs) for patient engagement. While each type has proven useful, they often function in isolation, addressing only part of patients' and clinicians' overall needs. The goal is to show that although the components for such a system already exist, combining them into a single, user-friendly, and autonomous platform remains an underexplored area with high potential, especially for improving healthcare accessibility in regions like Oman.

Telepresence robots, often described as a mobile screen, are widely used for remote medical consultations. Systems such as InTouch Health RP-VITA and the VGo robot have been used successfully in "telestroke" care, allowing neurologists to assess patients remotely, reduce diagnosis time, and improve outcomes [5]. These systems support remote rounds, specialist consultations, and family visits, cutting travel costs and reducing infection risks. However, studies also note several drawbacks. A review by Kristoffersson et al. found that telepresence robots, though functional, often feel impersonal and difficult to control [6]. Clinicians report high cognitive load when navigating cluttered hospital spaces. Patients, especially the elderly and children, also find these robots less engaging than in-person visits due to limited physical embodiment and lack of expressive cues [7]. This "presence gap" limits their effectiveness beyond basic video conferencing, reducing their ability to offer comfort and reassurance during care.

This literature review aims to build a comprehensive case for the development of an integrated, autonomous healthcare robotics platform. It begins by examining the current state and limitations of distinct domains within healthcare robotics: telepresence for remote consultation, socially assistive robots for patient engagement, and autonomous mobile robots for logistics. The review then draws parallels with the consumer and industrial sectors, where autonomous navigation has reached maturity and widespread adoption. By systematically analyzing the strengths and weaknesses of these disparate technologies, this survey identifies a critical research gap: the absence of a unified system that synthesizes clinical telepresence, social interaction, and autonomous mobility. The purpose is to establish a clear rationale for a novel robotic solution that merges these capabilities to create a more holistic and effective tool for patient care.

2.1.1 Success Stories: Consumer, Industrial, and Quadruped Robotics

1. **Consumer Market: The Robotic Vacuum Revolution.**
   The most successful deployment of autonomous mobile robots to the general public has been robotic vacuum cleaners, which dominate the consumer robotics market without serious competition from traditional cleaning methods. The global robotic vacuum cleaner market has experienced explosive growth, with projections indicating expansion from $9.07 billion in 2024 to $31.79 billion by 2033, representing a compound annual growth rate (CAGR) of 16.97% [8]. Global shipments reached 15.35 million units in the first half of recent years, demonstrating a 33% year-over-year growth [9].

Figure 1: Modern robotic vacuum cleaners utilize SLAM and localization technologies to autonomously navigate residential spaces, representing the most successful consumer robotics application to date.

The key to their success lies in the sophisticated implementation of SLAM (Simultaneous Localization and Mapping) and localization technologies. Modern robotic vacuum cleaners employ LiDAR sensors (Light Detection and Ranging), visual SLAM (VSLAM), and advanced mapping algorithms to create real-time maps of residential environments, enabling efficient path planning and obstacle avoidance [10]. This technology has proven so reliable that robotic vacuums have achieved nearly 40% of the service robot market share [11], demonstrating that autonomous navigation, when properly implemented, is mature enough for widespread consumer adoption.

2. **Industrial Market: Amazon's Warehouse Robotics Dominance.**
In the industrial sector, the most successful large-scale application of AMRs (Autonomous Mobile Robots) has been warehouse automation, with Amazon leading the field. Amazon reports more than 750,000 to over one million robots across its fulfillment network, including drive units, mobile manipulators, and next-generation platforms such as Pegasus and Xanthus [12, 13]. These fleets demonstrate reliable fleet management, safe human–robot collaboration, and 24/7 uptime in highly dynamic environments.

Figure 2: Amazon Pegasus drive unit used in high-throughput fulfillment centers [12].

3. **Quadruped Robotics: Unitree Go2 as a Robust Mobility Platform.**
State-of-the-art quadruped robots such as Unitree Go2 demonstrate robust mobility on unpredictable terrain using reinforcement learning for locomotion policy optimization, advanced multi-sensor perception (3D LiDAR + depth/RGB cameras), and high-performance

onboard compute for real-time planning and control. These capabilities enable operation on stairs, slopes, and uneven surfaces, with compelling applications in inspection, security, and emergency response [14, 15]. Although not our primary target platform, Unitree's capabilities illustrate complementary directions for rugged environments using RL and advanced multi-sensor perception, and inspire future extensions.

Figure 3: Unitree Go2: quadruped platform featuring RL locomotion, multi-sensor perception, and high-performance compute [14].

2.1.2 Benefits of Autonomous Mobile Robots in Healthcare Settings

Recent literature and deployment data reveal multiple evidence-based benefits of AMRs in hospital environments:

Infection Control and Reduced Human Contact. AMRs significantly minimize human-to-human contact, reducing cross-contamination risks a critical advantage demonstrated during the COVID-19 pandemic [16, 17]. Autonomous delivery robots eliminate the need for multiple staff members to handle potentially contaminated materials, reducing occupational exposure and conserving personal protective equipment. UV-C (ultraviolet-C) disinfection robots have been shown to achieve 99.9% pathogen kill rates on environmental surfaces, helping reduce hospital-acquired infections (HAIs) [18].

Wayfinding and Visitor Guidance. Intelligent hospital guidance robots provide accurate navigation assistance to visitors and patients, reducing confusion and staff burden [19]. These robots can escort visitors to patient rooms, provide multilingual directions, and answer frequently asked questions about hospital facilities, allowing staff to focus on clinical care.

Material Transport and Delivery. AMRs excel at routine delivery tasks including medications, meals, linens, laboratory specimens, and medical supplies. This automation frees nursing staff from non-clinical transport duties, allowing them to spend more time on direct patient care. Studies indicate that delivery robots can reduce staff time spent on logistics by up to 30% [20].

Psychological Support and Patient Morale. Research consistently demonstrates that pediatric patients exhibit positive emotional responses to robot interactions [21]. Socially interactive robots reduce anxiety, provide distraction during painful procedures, and improve treatment compliance in children. Studies show that patients perceive robotic companions as non-judgemental, reducing psychological barriers to seeking help or expressing concerns [22].

Autonomous Disinfection and Cleaning. Cleaning and disinfection robots can be equipped with UV-C light systems at relatively low cost, providing continuous environmental sanitation [16]. These robots operate during off-peak hours without human supervision, ensuring consistent hygiene standards while minimizing disruption to clinical workflows.

Telepresence and Virtual Presence. AMRs with integrated telepresence capabilities enable remote consultations, family visits, and specialist rounds without requiring travel [5]. This technology proved invaluable during pandemic restrictions and continues to provide value for rural healthcare access and specialist consultations.

Waste Collection and Handling. Some advanced AMR systems are deployed for biohazardous waste collection and disposal, reducing staff exposure to potentially infectious materials. In China and other Asian markets, robots have been successfully implemented for collecting and transporting medical waste from patient rooms to centralized disposal areas.

2.1.3 Challenges in Hospital AMR Deployment

Despite the significant benefits, several challenges remain:

Cost and Return on Investment. Hospital-grade AMRs represent substantial capital investments, typically ranging from $50,000 to $150,000 per unit depending on capabilities. Healthcare administrators require clear Return on Investment (ROI () demonstration through labor savings, efficiency gains, and improved outcomes to justify procurement [20].

System Integration Complexity. Integrating AMRs with existing hospital information systems (HIS), electronic health records (EHR), elevator controls, automatic doors, and staff communication systems presents significant technical challenges. Interoperability standards are still evolving, requiring custom integration work for each deployment [17].

Safety, Privacy, and Trust. Regulatory compliance (HIPAA, GDPR), patient data security, and physical safety in crowded environments remain critical concerns. Healthcare facilities must ensure that robots can safely navigate around patients, visitors, and medical equipment while maintaining patient privacy and data security [23]. Building trust among staff and patients requires careful change management and demonstration of reliability.

Regulatory and Liability Issues. Medical device regulations, liability frameworks for autonomous systems, and insurance considerations create additional barriers to widespread adoption. Healthcare organizations must navigate complex approval processes and establish clear accountability frameworks for robot operations.

2.1.4 Overview of Existing Robots in Healthcare

The hospital robotics market has matured significantly, with several manufacturers deploying autonomous mobile robots at scale. Table 2 presents a comprehensive comparison of leading hospital robot platforms currently deployed in eastern markets. Note that prices are approximate estimates and can vary significantly depending on region, configuration, volume, and additional features.

Rank	Manufacturer & Model	Est. Deployments	Est. Price (USD)	Key Hospital Applications	Features & Notes
1	Pudu Robotics – PuduBot 2 [24]	15,000–20,000 units (5,000 healthcare)	$12,000	Medication/meal delivery, linen transport, waste collection	Multi-floor navigation (elevator integration), VSLAM+LiDAR, 40 kg payload, 6–8 hr battery. Deployed in 100+ hospitals globally (China, Singapore, U.S.); customizable trays, infection-resistant coating.
2	Keenon Robotics – M2 Medical Robot [25]	8,000–10,000 units (healthcare-focused)	$47,500	UV-C disinfection, medication/lab sample delivery, patient room cleaning	UV sterilization (99.9% pathogen kill rate), 40 kg payload, voice interaction, 10 hr runtime. Used in 100+ Chinese hospitals during COVID; exported to U.S./Europe (HIPAA-compliant).
3	Pudu Robotics – PUDU CC1 [26]	5,000–7,000 units (healthcare + hospitality)	$16,800	Floor cleaning, disinfection, material transport	Autonomous scrubbing (1,500 m^2/hr), self-charging, obstacle avoidance. Popular in Chinese/U.S. hospitals for large-scale sanitation; integrates with PuduBot for hybrid tasks.
4	Keenon Robotics – Peanut Medical [27]	3,000–5,000 units	$12,700	Medication/meal delivery, patient guidance, telepresence	Compact (59 cm wide), multi-modal navigation, 30 kg payload. Deployed in Asia-Pacific hospitals; supports remote doctor-patient interaction via screen.

Table 2: Competitive analysis of leading hospital AMR platforms by global deployment and capabilities.

Figure 4: Pudu Robotics PuduBot 2

Figure 5: Keenon M2 Medical Robot

Figure 6: Pudu Robotics PUDU CC1

Figure 7: Keenon Peanut Medical

In parallel, the field of Socially Assistive Robotics (SAR) has focused on the psycho-social aspects of patient care. These robots are designed not to perform physical tasks but to engage, encourage, and comfort patients through social interaction. For pediatric patients, robots like NAO and PARO (a therapeutic baby seal robot) have been shown to significantly reduce pain perception and anxiety during painful medical procedures such as vaccinations and chemotherapy [21]. The robot acts as a coach and a distractor, creating a more positive and cooperative atmosphere. For elderly populations, SARs have been explored for companionship to combat loneliness, provide cognitive stimulation for dementia patients, and deliver medication reminders to improve treatment adherence [28]. These studies confirm that a robot's ability to interact socially is a powerful tool for improving patient well-being and clinical outcomes. The primary limitation of SARs, however, is that they are typically specialized, non-mobile or semi-autonomous platforms that lack the high-fidelity telepresence capabilities needed for direct clinical consultation.

2.2 Technical Building Blocks

This section deconstructs the essential technologies that form the foundation of an advanced autonomous healthcare robot. It explores supplementary localization methods, including Radio Frequency (RF) Technologies like UWB and AoA, followed by a deeper dive into SLAM-Based Sensor Fusion Techniques that integrate multiple data streams for robust performance. Finally, it examines the role of Conversational AI and Large Language Models (LLMs), which provide the intelligence for human-robot interaction, and explores User Interface (UI) and Human-Robot Interaction (HRI) paradigms for effective clinical integration.

2.2.1 Navigation System

The challenge of manual navigation highlighted in telepresence research is being addressed by autonomous mobile robots (AMRs). Modern AMRs employ a combination of SLAM, path planning, and obstacle avoidance to navigate complex environments without human intervention. The core navigation stack typically includes Radio Frequency (RF) Localization Technologies and SLAM-Based Sensor Fusion Techniques.

Radio Frequency (RF) Localization Technologies Radio frequency (RF) refers to electromagnetic waves used for wireless communication (roughly 3 kHz–300 GHz). Works where GPS cannot provide reliable positioning, such as indoor coverage through walls/obstacles and large areas. Various RF technologies including Wi-Fi, RFID, Bluetooth Low Energy (BLE), ZigBee, and Ultra-Wideband (UWB) are employed, with selection criteria based on coverage, scalability, battery life, accuracy, and cost. UWB solutions offer extended coverage and strong scalability due to multiple nodes sharing spectrum. UWB tags typically achieve multi-year battery life. Cost is somewhat higher; for example, instrumenting a 5000m^2 facility may cost up to $20,000. Reported accuracy in industrial scenarios varies (approximately 20cm to greater than 1 m) and degrades in cluttered or metallic environments or very large warehouses [29]. The general structure of these systems is to use static reference sensor locations, or anchors, within an environment, along with beacons or tags fixed to the objects with desired tracking. The location of the tracked object can be estimated using readings from the anchors. Mathematically, this estimation is commonly called multilateration (also called trilateration) when using distance measurements.

Figure 8: UWB anchor–tag geometry for ranging and multilateration-based localization in indoor environments.

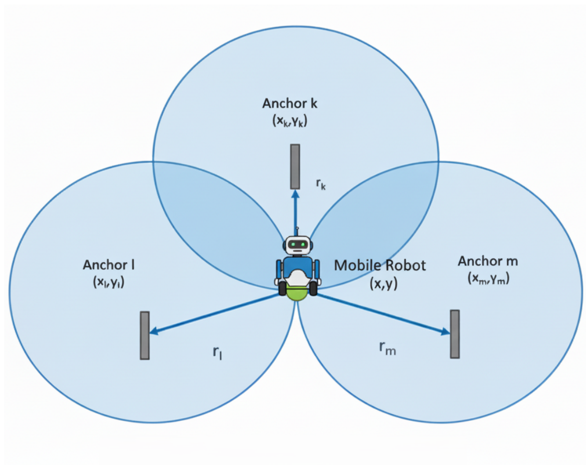

Figure 9: Illustration of multilateration technique. Each known anchor forms a circle with radius equal to its distance to the robot, and the robot's estimated position is where these circles intersect.

AoA measures the direction (bearing) from which a radio signal arrives at a receiver. An antenna array senses time differences across elements and apply signal-processing algorithms such MUSIC or ESPRIT convert those differences into an angle. AoA (often via BLE 5.1+ or UWB Direction Finding) estimates a target's angle using a beacon or tag and a robot-mounted antenna array, reducing reliance on dense static infrastructure. BLE-based tags offer excellent battery life (years). Limitations include line-of-sight sensitivity and variable accuracy; hybrid AoA + RSSI filtering can improve precision, achieving mean errors on the order of 12 cm in simulation even in cluttered settings [30]. AoA could be used to implement a soultion that doesnt count on localization or positioning infrastructure in the environment, instead having the robot home in on a beacon worn by the user. This would be similar to how some Bluetooth trackers work for finding lost items. The robot would use the angle of arrival data to determine the direction to move towards the beacon signal, allowing it to autonomously navigate to the user's location. This approach could be particularly useful in dynamic environments like hospitals where fixed infrastructure may not be feasible. Figure 10 illustrates a proposed setup where a Bluetooth homing controller on the robot uses AoA data from a wearable beacon to guide navigation towards the target. Multiable anchors could be used to implement waypoints follwing navigation paths through the environment.

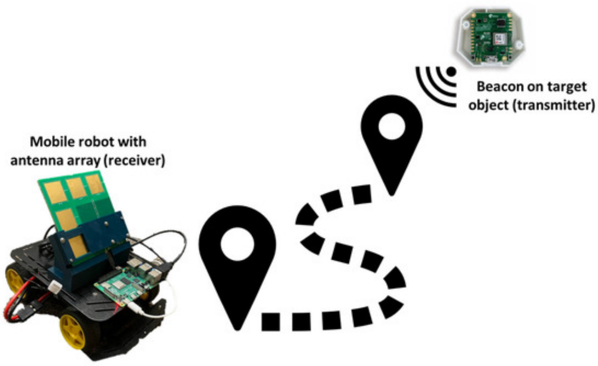

Figure 10: Configuration of proposed setup of Bluetooth homing controller.

SLAM-Based Sensor Fusion Techniques Simultaneous Localization and Mapping (SLAM) is a critical technology for autonomous navigation, enabling robots to build maps of unknown environments while simultaneously keeping track of their own location. Its effectiveness hinges on robust sensor fusion techniques that integrate data from various sources such as camera(s), IMU, wheel odometry, or RF ranging . Beyond external RF systems, autonomous navigation heavily relies on SLAM with multi-sensor fusion for robustness. A common stack fuses 2D LiDAR with an IMU and wheel odometry. IMU and odometry constrain motion, correcting LiDAR motion distortion and mitigating slip-induced drift. Studies report substantial gains when integrating IMU and wheel odometry, for example, approximately 27% improvement in rotation error and greater than 67% reduction in position error relative to LiDAR only SLAM, at the expense of added system complexity and compute [31].

VSLAM (Visual Simultaneous Localization and Mapping) is a method that uses cameras to help robots figure out where they are and build maps of their surroundings at the same time. VSLAM uses cameras like monocular (single), stereo (two), or RGB-D (depth-sensing) for positioning and mapping. It provides low-cost, lightweight hardware and better scene understanding. Challenges include poor performance in texture-less or repetitive areas, and scale issues in monocular systems (e.g., early ORB-SLAM 2.0). Modern approaches prefer hybrid Visual–Inertial–Ranging–LiDAR (VIRAL) SLAM for improved reliability in complex environments [32]. Figure 11 shows the combination of visual/laser data with inertial and odometric inputs for stable estimation. Tools like Google Cartographer build 3D maps in real-time, using LiDAR, IMU, and odometry.

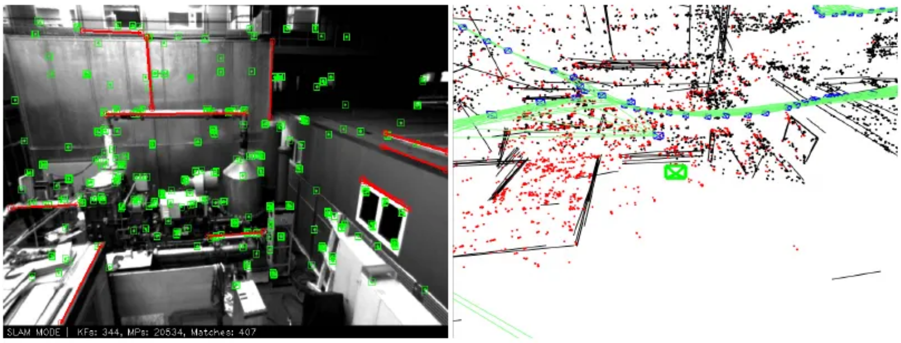

Figure 11: Illustration of SLAM fusion: visual/laser perception integrated with inertial and odometric constraints for robust state estimation.

Navigation Systems Across Hospital Robotics Platforms. Table 3 summarizes the navigation technologies and sensor configurations employed by leading hospital robotics platforms mentioned in table 2.

Robot Model	**Primary Localization**	**Mapping Technology**	**Key Sensors**
PuduBot 2 [24]	SLAM (LiDAR-based)	Real-time 2D/3D mapping	2D LiDAR, depth camera, IMU
M2 Medical Robot [25]	SLAM (LiDAR-based)	Real-time 2D mapping	2D LiDAR, RGB camera, encoders
PUDU CC1 [26]	SLAM (LiDAR-based)	Real-time 2D mapping	2D LiDAR, ultrasonic sensors, encoders
Peanut Medical [27]	Hybrid (SLAM + RF)	Real-time 2D mapping	LiDAR, depth sensors, optional UWB

Table 3: Navigation systems and sensor technologies across leading hospital robotics platforms.

2.2.2 Mobile Robot Platform specifications

The selection of an appropriate mobile robot platform is a critical decision that directly impacts system reliability, development time, and operational costs. For a healthcare telepresence robot operating in hospital corridors and patient rooms, the platform must satisfy several essential technical and operational requirements while remaining within practical budget constraints.

Platform Requirements. The system requires a wheeled mobile robot platform with the following capabilities:

- **Autonomous docking and charging**: The platform must support autonomous return-to-base functionality with a charging dock, enabling unattended operation and automatic battery management. High-level API commands such as `dock()` or `return_to_charge()` are essential for seamless integration with the navigation stack.
- **ROS/ROS 2 compatibility**: Native support for the Robot Operating System is required to leverage existing navigation packages (Nav2, SLAM Toolbox, AMCL) and facilitate rapid development and integration of custom behaviors.
- **Payload capacity**: The platform must support 5–10 kg of additional payload to accommodate a telepresence screen (tablet or monitor), onboard computer (Raspberry Pi or Jetson Nano), camera system, and potential accessories such as speakers or mounting hardware.

- **Indoor navigation sensors**: Integration points for LiDAR (2D or 3D), depth cameras, or other sensors required for SLAM-based autonomous navigation in dynamic hospital environments.

- **Safety and reliability**: Cliff detection, bumper sensors, emergency stop functionality, and predictable motion control to ensure safe operation around patients, visitors, and medical staff.

Project Constraints. Budget constraints significantly influence platform selection. The target budget for the mobile base, charging dock, and core sensors is approximately $1,000–$1,500 USD, recognizing that higher-cost industrial platforms ($5,000+) exceed the scope of an educational research project. Additional constraints include global availability (shipping to Oman), vendor support and documentation quality, and community resources for troubleshooting and development.

Platform Comparison. Table 4 presents a comparative analysis of commercially available mobile robot platforms.

Platform	Price (USD)	Dock	ROS	Payload	Notes
Budget Tier ($500–$1,000)					
iRobot Create 3 [33]	$549	Yes	ROS 2	9 kg	Roomba-based, proven docking, ideal for education
Yahboom ROSMASTER X3 [34]	$699	DIY	ROS 1	5–8 kg	Mecanum wheels, requires custom docking solution
Elephant Robotics MyAGV 2023 [35]	$899	DIY	ROS 1	8 kg	Mecanum wheels, 2D SLAM ready, Python/ROS compatible
Mid Tier ($1,000–$2,000)					
SLAMTEC Apollo [36]	$1,200	Yes	ROS 1/2	50 kg	Same platform as Pudu/Keenon hospital robots
TurtleBot 4 Standard [37]	$1,400	Yes	ROS 2	9 kg	Research-grade, Nav2 pre-configured, Lidar included
Husarion ROSbot 2R [38]	$1,600	Optional	ROS 2	10 kg	Mecanum wheels, hospital navigation demos available
AgileX LIMO Pro [39]	$1,799	Optional	ROS 2	10 kg	4-mode steering, Jetson Nano, research platform

Table 4: Comparative analysis of mobile robot platforms for healthcare telepresence applications.

Platforms Table Discussion. Table 4 provides a compact comparison of candidate mobile bases across price tiers, emphasising the primary trade-offs faced during platform selection: cost versus out-of-the-box capabilities (autonomous docking, ROS/ROS2 support, payload capacity, and built-in sensors). Budget platforms offer low acquisition cost and strong community resources but typically require additional engineering (docking solutions, sensor integration). Mid-tier platforms reduce integration risk by providing native docking and ROS support at higher cost.

2.2.3 Conversational AI and Large Language Models (LLMs)

The field of large language models (LLMs) has witnessed transformative advancements, establishing them as a core technology for conversational AI. An LLM is a sophisticated artificial intelligence system trained on vast datasets of text and code, enabling it to understand, generate, and interact with human language in a nuanced manner. These models have rapidly transitioned from research prototypes to industrial-scale technologies, resulting in a proliferation of models with distinct strengths and commercial terms.

The LLM landscape is broadly divided into two categories: open-source and proprietary. Open-source models, such as Meta's Llama series and the DeepSeek series, offer significant flexibility, allowing for deep customization and fine-tuning to meet specific application needs, often at a lower cost [40]. However, they may require more complex setup and can sometimes lag behind proprietary counterparts in raw performance. In contrast, privately-owned models like OpenAI's GPT series and Google's Gemini series typically offer state-of-the-art performance and are adept at handling highly complex tasks. This performance comes with trade-offs, including higher operational costs and potential data privacy concerns, which are critical considerations in healthcare applications [41]. The selection of an LLM depends on a careful balance between performance requirements, budget, customizability, and data governance policies. The following table provides a consolidated overview of the key characteristics for each of those 4 LLMs.

Model (Version)	**Org.**	**License**	**Max Context (tokens)**	**Input API Cost ($/M)**	**Output API Cost ($/M)**
OpenAI GPT-4o	OpenAI	Proprietary	128,000	$2.50	$10.00
Google Gemini 2.5 Pro	Google	Proprietary	1,048,576	$1.25–$2.50	$10.00–$15.00
Meta Llama 3.1 405B	Meta	Llama 3.1 Community License	128,000–131,072	~$0.80–$3.50	~$0.80–$3.50
DeepSeek V3	DeepSeek	MIT (Code) / DeepSeek Model	128,000–164,000	~$0.14–$0.48	~$0.28–$1.10

Table 5: Comparative analysis of leading Large Language Models. Data compiled from early 2025 period [40].

LLM Discussion and Cost Considerations. Table 5 summarizes key characteristics of leading LLM options. OpenAI's GPT-4o offers strong performance with a 128k token context window, while Google's Gemini 2.5 Pro stands out with exceptional context length (over 1M tokens) and competitive API pricing. Notably, Google's Gemini API is available at no cost for requests under 60 requests per minute (RPM), making it an attractive option for development and testing phases. Meta's Llama 3.1 405B provides an open-source alternative with flexible licensing, while DeepSeek V3 offers the most economical API costs, particularly beneficial for cost-sensitive deployments. Even that llama and DeepSeek are open-source models, they still have associated costs when using hosted APIs, which can vary based on usage volume and specific service providers.

2.2.4 User Interface and Human-Robot Interaction

For autonomous healthcare robots operating in clinical environments, the user interface (UI) and human-robot interaction (HRI) design directly impacts adoption, user satisfaction, and clinical workflow integration. The interface must balance accessibility for diverse user groups (clinicians, administrative staff, patients) with robust control mechanisms suitable for dynamic hospital environments. Common UI paradigms employed in hospital robotics include:

Onboard Touchscreen Interface. A dedicated touchscreen mounted on the robot body provides direct, localized control and real-time status feedback. Common in systems like Pudu Robotics' PuduBot 2 and Keenon Robotics' M2 Medical Robot, these interfaces typically run Android OS and display navigation status, task queues, and emergency controls. Advantages include intuitive interaction without external devices and immediate feedback. Limitations include reliance on the robot's physical proximity and potential hygiene concerns in sterile environments [25].

Mobile Application (iOS/Android). Native mobile applications running on clinicians' and administrators' smartphones or tablets enable remote operation and fleet monitoring from anywhere within hospital networks. This approach is ubiquitous across commercial systems (PuduBot 2, M2, Peanut Medical) and supports features such as route customization, real-time tracking, task scheduling, and status notifications. Mobile apps offer flexibility and integrate seamlessly with existing hospital workflows, though they require network connectivity and depend on user device management [24].

Web-Based Dashboard. Cloud-hosted or on-premises web portals provide centralized fleet management, historical analytics, and administrative controls for hospital operators and IT administrators. Systems like Keenon Robotics and Pudu Robotics include web dashboards for route optimization, performance metrics, and multi-robot coordination. Web interfaces support remote monitoring across multiple locations and facilitate integration with hospital information systems (HIS). Trade-offs include network dependency, data security considerations, and potential latency in real-time control [26].

Voice Control and Natural Language Interfaces. Voice-based interaction leveraging conversational AI and LLMs (as discussed in Section 2.2.3) represents an emerging paradigm for hands-free operation in clinical settings. Systems like Keenon's M2 support voice commands, allowing clinicians to interact with robots while maintaining hygiene and workflow continuity. Voice interfaces reduce cognitive load and support more natural, intuitive interactions, but require robust speech recognition, noise handling (hospital environments are loud), and contextual understanding [41].

Hospital Integration via APIs. Integration with existing hospital IT infrastructure through RESTful APIs or DICOM/HL7 standards allows robots to receive task assignments directly from electronic health records (EHR) or hospital management systems. This approach minimizes manual input and ensures tasks are coordinated with clinical workflows. Examples include systems capable of receiving delivery requests directly from hospital pharmacies or labs [26].

Robot UI Usage Across Hospital Platforms. Table 6 summarizes the UI and interaction modalities employed by leading hospital robotics platforms, illustrating the diversity of approaches and trade-offs in current systems.

Robot Model	**Onboard Screen**	**Mobile App**	**Web Dashboard**	**Voice/AI Interface**
PuduBot 2 [24]	Android tablet	iOS/Android	Yes	Limited
M2 Medical Robot [25]	Android display	iOS/Android	Yes	Yes (voice control)
PUDU CC1 [26]	Android panel	iOS/Android	Yes	No
Peanut Medical [27]	Optional	iOS/Android	Yes	Limited

Table 6: UI and interaction modalities across leading hospital robotics platforms.

2.3 Conclusion

The literature presents a clear picture:

1. **Telepresence robots** are effective for remote diagnosis but are limited by manual control and an impersonal user experience.

2. **Socially assistive robots** have proven their value in patient engagement and comfort but lack clinical telepresence and advanced mobility.

3. **Autonomous mobile robots** have mastered navigation in hospitals but are used for logistics, not patient interaction.

4. **Conversational AI** offers intelligent dialogue but lacks the physical embodiment needed for meaningful healthcare encounters.

5. **Consumer and industrial robotics** have demonstrated that autonomous navigation technology is mature, reliable, and cost-effective for real-world deployment, as evidenced by millions of robotic vacuum cleaners in homes and over one million AMRs in Amazon warehouses.

6. **Current hospital AMRs** are predominantly focused on material transport and disinfection, with limited integration of social interaction, advanced telepresence, and conversational AI capabilities.

The critical research gap lies at the intersection of these domains. There is a demonstrable need for a single, integrated platform that combines the **clinical access** of a telepresence robot, the **patient-centric engagement** of a SAR, the **operational independence** of an autonomous navigator, and the **interactive intelligence** of modern conversational AI. The proven success of autonomous navigation in consumer (robotic vacuums) and industrial (Amazon warehouses) applications demonstrates that the core technology is mature and ready for advanced healthcare applications.

This project directly addresses this gap by designing a holistic system that synthesizes these strengths, creating a smart healthcare robot capable of not only connecting doctors and patients but also navigating its environment independently and engaging with patients in a helpful and empathetic manner. By learning from the successes of consumer robotics, warehouse automation, and existing hospital AMR deployments, this project aims to advance the state-of-the-art in healthcare robotics toward truly integrated, intelligent, and patient-centered care.

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